JAVA Training Syllabus

Core Java

Introduction & Setup

- What is Java and why it is used
- Java versions and evolution
- JDK, JRE, JVM, JIT architecture
- Writing, compiling, and running Java programs
- Editors and IDE basics (Eclipse)
- Errors: syntax, runtime, exceptions

Java Fundamentals

- Naming conventions, keywords, literals
- Data types (primitive & reference)
- Variables, type casting
- Operators
- Statements, blocks, comments
- Interview-based coding questions

Control Structures

- Decision making: if, else, switch
- Loops: for, while, do-while
- Problem-solving using loops

Input & Arrays

- Scanner and BufferedReader
- Arrays (1D, 2D)
- Array operations (sorting, resizing concept)
- String vs char array
- Common interview problems

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Object-Oriented Programming (OOP)

- Classes and objects
- Constructors (types, overloading)
- this keyword
- Static vs instance members
- Encapsulation, inheritance, polymorphism, abstraction
- Access modifiers
- Method overloading vs overriding
- Call by value
- Object creation patterns

Advanced Core Concepts

- Command-line arguments
- Inner classes
- Packages and classpath
- Java API (String, Math, Wrapper classes)
- Garbage collection
- Enum, auto-boxing
- JAR files

Exception Handling

- try-catch-finally
- throw, throws
- custom exceptions
- exception hierarchy
- checked vs unchecked exceptions

Multithreading

- Thread lifecycle
- Thread vs Runnable
- Synchronization
- Deadlock
- Inter-thread communication
- Executor framework basics

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I/O Streams

- File handling
- Byte vs character streams
- BufferedReader, FileReader, etc.
- Serialization & deserialization

Collections Framework

- List, Set, Map
- ArrayList, LinkedList, HashMap, TreeMap
- Iterator, Comparator
- Queue, Deque
- Generics
- Sorting using Collections

Java 8 Features (Intro)

- Lambda expressions
- Streams API (basic idea)
- Functional interfaces

Advanced Java

JDBC (Database Connectivity)

- Connecting Java with database
- Driver types
- Statements, PreparedStatement
- ResultSet
- Transactions
- Batch updates

Web Basics

JSP (Java Server Pages)

- JSP lifecycle
- Scriptlets, expressions, directives
- Implicit objects
- MVC architecture

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- JSTL and custom tags
- CRUD applications

Servlets

- Servlet lifecycle
- HttpServlet
- Request & response handling
- Session tracking (cookies, session, URL rewriting)
- RequestDispatcher, redirect
- Deployment on Tomcat
- Real-time web applications

Concurrency & Collections Implementation

- Custom implementations of List/Map
- Executor service
- Multi-threaded file processing

JavaMail API

- Sending emails via Gmail
- HTML emails

Design Patterns

- Singleton
- Factory
- DAO / DTO
- MVC
- Front Controller

Testing & Build Tools

- JUnit testing
- Maven project management
- pom.xml basics
- Dependency management

Project Work

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